Additional Exercises 1

Positive Series — Convergence Tests

Mirrors: SIT1 | ★ Easy ★★ Medium ★★★ Hard

Exercise 1: Determine convergence and find the sum

Use geometric series $\sum_{n=0}^{\infty} ar^n = a/(1-r)$ for $|r| < 1$, or telescoping where applicable.

(a) ★ $\displaystyle\sum_{n=1}^{\infty} \left(\frac{3}{4}\right)^n$

(b) ★ $\displaystyle\sum_{n=0}^{\infty} \frac{2^n + 3^n}{5^n}$

(c) ★ $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$ *(telescoping)*

(d) ★★ $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n(n+2)}$ *(partial fractions)*

(e) ★★ $\displaystyle\sum_{n=1}^{\infty} \left[\sin\frac{1}{n} - \sin\frac{1}{n+1}\right]$

(f) ★★ $\displaystyle\sum_{n=1}^{\infty} \ln\left(1 - \frac{1}{n^2}\right)$ *(write as telescoping using $\ln\frac{n-1}{n} + \ln\frac{n+1}{n}$)*

(g) ★★★ $\displaystyle\sum_{n=1}^{\infty} \frac{4^{n+1}}{5^n}$

(h) ★★★ $\displaystyle\sum_{n=1}^{\infty} \frac{1}{(3n+1)(3n+4)}$

Exercise 2: Determine convergence or divergence (no sum needed)

State which test you use and why.

(a) ★ $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n^2+1}$

(b) ★ $\displaystyle\sum_{n=1}^{\infty} \frac{n}{2^n}$

(c) ★ $\displaystyle\sum_{n=1}^{\infty} \frac{n^2}{n!}$

(d) ★ $\displaystyle\sum_{n=2}^{\infty} \frac{1}{n(\ln n)^2}$

(e) ★★ $\displaystyle\sum_{n=1}^{\infty} \frac{2^{1/n}}{3^n}$

(f) ★★ $\displaystyle\sum_{n=1}^{\infty} \frac{\arctan(1/n)}{n}$

(g) ★★ $\displaystyle\sum_{n=1}^{\infty} \frac{1 \cdot 3 \cdot 5 \cdots (2n-1)}{(2n)!}$

(h) ★★ $\displaystyle\sum_{n=1}^{\infty} \left(\frac{n}{3n-1}\right)^{2n-1}$

(i) ★★★ $\displaystyle\sum_{n=1}^{\infty} \left(\frac{n+1}{2n-1}\right)^{n}$

(j) ★★★ $\displaystyle\sum_{n=1}^{\infty} \frac{n!}{1 \cdot 3 \cdot 5 \cdots (2n-1)}$

(k) ★★★ $\displaystyle\sum_{n=1}^{\infty} \frac{(2n)!}{4^n (n!)^2}$

Exercise 3: Necessary condition & basic properties

(a) ★ State the necessary condition for convergence. Use it to immediately decide on: $\displaystyle\sum_{n=1}^{\infty} \frac{n^2}{n^2+1}$ and $\displaystyle\sum_{n=1}^{\infty} (-1)^n$.

(b) ★★ Prove: if $\sum a_n$ converges and $c > 0$, then $\sum c a_n$ converges with sum $c \sum a_n$.

(c) ★★ Prove: if $\sum a_n$ converges and $\sum b_n$ diverges, then $\sum (a_n + b_n)$ diverges.

(d) ★★★ A ball is dropped from height $h = 4$ m and bounces to $2/3$ of its previous height. Find the total distance traveled.

(e) ★★★ For which values of x does $\displaystyle\sum_{n=1}^{\infty} \tan^n(x)$ converge? Find the sum where it converges.

Exercise 4: Integral test

The series $\sum f(n)$ *and* $\int_1^\infty f(x)\,dx$ *have the same convergence behaviour when* f *is positive, continuous, and decreasing.*

(a) ★ $\displaystyle\sum_{n=2}^{\infty} \frac{1}{n \ln n}$

(b) ★★ $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n^2+1}$ *(use* $\int 1/(x^2+1)\,dx = \arctan x$*)*

(c) ★★★ $\displaystyle\sum_{n=1}^{\infty} n\,e^{-n^2}$

★ *Direct geometric or p-series comparison* ★★ *Ratio, root, or limit comparison test* ★★★ *Combination of tests or subtle argument required.*